NAGALAND LITERACY AND NUMERACY FEST 2026

Micro-Improvement Project (Science Teacher) — Grade 8

Chapter 9 — The Magic of Science: Solutes, Solvents & Solutions

Why Micro-Improvement Project (MIP)?

In this project, learners turn into science magicians by designing quirky, hands-on tricks that reveal the ideas of solutes, solvents, solutions and density. Groups can make bottles with colourful liquid layers, a set-up where objects mysteriously float or sink, or other illusion-like demonstrations of their own. Each group explains the science behind its trick and makes an eye-catching poster for it. On 'Science Magic Show' day, learners set up stalls like the ones at the Hornbill Festival at Kisama and perform their tricks for the village revealing the science that makes the magic happen.

Do upload pictures of the process — such as the group work while making the tricks and the final products — to the project completion form.

Leading Question: How can we turn science into magic that surprises our village?

Chapter Covered	Chapter 9: The Amazing World of Solutes, Solvents and Solutions
Total Time Required	1 Week 40 minutes a day for 5 days.
Resources Required	For a class, Teachers may form groups based on the total number of students in their class. Ideally, each group should have around **3-4 students**, but the group size and number of groups can be adjusted depending on class enrolment. Across the 5 days: 8–10 glasses or steel tumblers and 1 bucket of water • about 1 kg salt (local salt is fine) and 250 g sugar • 8 eggs, 8 lemons, 1 small bottle of oil, and a little honey or thick syrup • 8 bamboo sticks, plus erasers and small stones from the surroundings • 8 chart papers, colours or chalk, and learners' own notebooks. Everything is freely available at home or costs only a few rupees per learner.

Learning Outcome

Knowledge-based Skills

- Identifies solutes, solvents and solutions in everyday mixtures.
- Sorts mixtures into uniform (solutions) and non-uniform mixtures.
- Explains concentration, saturation and solubility, and the factors (such as temperature) that affect them.
- Measures and reasons about density using simple, locally available tools and objects.
- Applies these science ideas creatively in an original demonstration, and explains the science clearly to an audience.

Key Concepts

- Demonstrates the concept of solute, solvent and solution using everyday local mixtures.
- Distinguishes saturated from unsaturated solutions, and explains the effect of temperature on solubility.
- Explains why objects float or sink using density (Density = Mass ÷ Volume).

> **Learning Competency C-1.1:** *"Classifies matter based on physical and chemical characteristics, including pure/impure and density."* **C-7.3:** *"Represents real-world events/relationships through diagrams and simple mathematical representations" (e.g. the solute–solvent tables and float/sink data tables in Tasks 1–3).*

Suggestions for Teachers

1. Read each day's plan and try the demonstrations yourself before class.
2. Form mixed, inclusive groups (gender and ability) before Task-1 and keep the same groups across all tasks.
3. Inform learners a day in advance when they need to bring materials from home, such as an egg or a bamboo stick.
4. Run this project before teaching Chapter 9, so learners discover the ideas through the tricks first.
5. Let salt, sugar and objects 'disappear' or 'float' first, and name the science only after learners have observed and guessed. Do not use taste or touch to identify anything.
6. Ask curiosity-building questions like 'How did that happen?' and encourage learners to think on their own.

Detailed Tasks

Task-1	The Bending Pencil — What Are Solute, Solvent and Solution?

Objective	Learners identify solutes, solvents and solutions in everyday mixtures, and set up their project by forming groups and agreeing on the leading question and project criteria.
Teacher's Note	Indoor. Classroom. Try the pencil-in-water demonstration yourself beforehand with your available glass and water.
Duration	40 minutes in class

Materials Needed

A transparent glass or tumbler, water, a pencil or thin bamboo stick, sugar, salt, a spoon, chart paper and chalk or a marker.

Activity Steps

1. Ask learners: ‘Have you ever seen a magic show, or a magician at a village fair or during the Hornbill Festival? Which trick do you remember? Have you ever tried a magic trick yourself?’ Take responses.
2. Demonstrate: fill a glass halfway with water, place a pencil or thin bamboo stick in it, and ask learners to look from the side. Ask what they see (the pencil looks bent or broken at the water’s surface) and whether it really bends.
3. Explain that this is refraction, a science idea studied later in the chapter on Light. Tell learners that in this project they will use solutes, solvents and solutions to build their own magic tricks and perform them at a Science Magic Show, with stalls like the ones at the Hornbill Festival. Share the leading question: ‘How can we turn science into magic that surprises our village?’
4. Divide learners into groups of 4–5 and assign roles: Leader, Recorder, Materials Manager, Reporter and Observer. Share the project criteria design at least one magic trick, using at least one solute/solvent/solution idea, with a scientifically correct explanation. Ask a volunteer to write the leading question and criteria on chart paper for the classroom wall.
5. Add one spoon of sugar to a glass of water, stir well, and ask where the sugar went, and where learners have seen a similar mixture at home (sweet tea, ORS, rice-washing water). Explain: a uniform mixture such as salt or sugar in water is called a solution; the dissolved solid is the solute and the liquid is the solvent Solute + Solvent = Solution.
6. In groups, learners fill in a solute–solvent table for local mixtures — ORS, salt water, lemon juice, instant coffee — using two worked examples (ORS: salt and glucose in water; salt water: salt in water) as a model. Take a few responses from each group and clear any doubts.

Homework Extension

Find 5 examples of solutions at home. For each, name the solute, the solvent and the solution.

Concept Built

A solution is a uniform mixture formed when a solute dissolves in a solvent (Solute + Solvent = Solution), identified through everyday local mixtures.

Task-2	The Disappearing Salt — Saturation and the Effect of Temperature

Objective	Learners distinguish saturated from unsaturated solutions, explain how temperature affects solubility, and begin planning their magic trick.
Teacher's Note	Indoor. Classroom.
Duration	40 minutes in class

Materials Needed

Two glasses or tumblers, cold water, hot water, salt, a spoon, notebook.

Activity Steps

1. Recap: a learner reads the leading question aloud; groups share their Day-1 homework and progress, and repeat the project criteria.
2. In groups, learners identify the solute and the solvent in more examples: tea, ORS, lemon water, vinegar, soda water, and soft drink.
3. Mix one spoon of salt into a glass of water and ask what happens to it and what will happen if more salt is added. Keep adding salt spoon by spoon, recording on the board after each spoon (up to about the seventh) whether it dissolves.
4. Ask what happens to the salt at first, and what learners notice near the end. Explain: a solution that can still dissolve more solute is unsaturated; a solution that cannot dissolve any more solute is saturated.

5. Ask whether the same salt in hot water would dissolve the same amount, less, or more. Repeat the activity with a glass of hot water, adding salt until it stops dissolving. Explain that for most substances, solubility increases with temperature so a saturated solution can behave like an unsaturated one once heated.
6. Leading question moment: in groups, discuss how the 'disappearing salt' or the hot-water change could become a magic trick that surprises the village. Groups prepare at least two trick ideas using solubility and/or temperature or example, a spoon of salt fully disappearing in water, or sugar dissolving visibly faster in hot water than in cold water placed side by side.

Concept Built

Saturated and unsaturated solutions, and the effect of temperature on solubility are the basis for the group's first magic trick ideas.

Task-3	Float or Sink? — Discovering Density

Objective	Learners explain why objects float or sink using density, and use this idea to improve their magic trick.
Teacher's Note	Indoor. Classroom. A large open vessel of water can also be managed in the schoolyard if that is easier.
Duration	40 minutes in class

Materials Needed

A bucket or large vessel of water, objects that float and sink (bamboo stick, metal spoon, plastic cap, eraser, small stone, egg), salt, notebook.

Activity Steps

1. Recap: read the leading question, share progress, and exchange notebooks to check Day-2 homework.
2. Show the objects — a bamboo stick, metal spoon, plastic cap, eraser and stone and ask learners to predict which will float and which will sink. Place the objects in the water and check. Ask which floated, which sank, which surprised them, and why a plastic cap floats but a metal spoon sinks.

3. Try another trick: place an egg in plain water in one glass and in salt water in another. Ask learners to predict where the egg will float and where it will sink in each, then test it.
4. Explain using a local example: a bamboo raft floats on the Doyang river while a stone dropped in the same river sinks. An object with higher density than water sinks, and an object with lower density than water floats. Density is the amount of mass in a given volume: Density = Mass ÷ Volume. Mass is the amount of matter in an object; volume is the space it takes up; and the density of a substance does not change with its shape or size.
5. Explain that the metal spoon sinks because its density is higher than water, and the plastic cap floats because its density is lower than water. If the same object is placed in different solutions, it floats in the one with higher density than itself, and sinks in the one with lower density which is why the egg behaves differently in plain water and salt water.
6. Learners predict, test, and record float/sink and the reason for a metal pin, a stone, a pencil or bamboo piece, a plastic ball, and a piece of apple.
7. Leading question moment: in groups, discuss how the idea of density brings the trick one step closer to surprising the village. Each group improves its trick using density — for example, an egg or eraser appearing to hover between layered liquids of different densities, or colourful liquids layered by density to look like they are floating. Each group names its trick, such as ‘Floating Egg Mystery’ or ‘Rainbow Water Bottle’.

Homework Extension

Bring the materials your group needs to test its trick on Day-4.

Concept Built

Objects float or sink based on their density relative to the liquid (Density = Mass ÷ Volume) — the second science idea behind the group’s magic trick.

Task-4	Testing, Feedback and the Poster

Objective	Learners test their magic trick, give and receive peer feedback, revise their trick, and prepare their presentation poster.
Teacher's Note	Indoor. Classroom, with groups spread out, or the yard if pouring liquids gets messy.
Duration	40 minutes in class

Materials Needed

Each group collected materials, water, salt, oil, honey or thick syrup, egg, chart paper and colours.

Activity Steps

1. Recap: read the leading question, share progress, repeat the project criteria, and exchange notebooks to check Day-3 homework.
2. Each group gathers its materials and tests its trick. Example 1, the layered-liquids trick: pour honey, water and oil slowly down the side of a spoon so the layers stay visible, pouring too fast mixes the layers. Example 2, the floating-egg trick: adjust how much salt is dissolved until the egg floats at the desired level; other objects such as a grape or a lemon wedge can also be tried for the same density effect.
3. Groups answer troubleshooting questions: Do the layers stay visible for at least one minute? Does the object float at the right level top, middle, or bottom?
4. Each group performs its trick for another group, which gives feedback using the Peer Feedback Form (Appendix 1) covering whether the trick works and is engaging, whether it uses a solution idea correctly, and how clearly the science is explained.
5. Leading question moment: each group discusses whether their trick, as it is now, would really surprise the village, and what would make it more surprising. Each learner writes down one thing they would change or improve; groups use this and the peer feedback to revise their trick.
6. Each group designs a poster with a catchy title (e.g. 'The Floating Egg Mystery' or 'Rainbow Water Bottle'), simple illustrated step-by-step instructions for the audience, and diagrams showing solutes dissolving in solvents and density differences such as honey, water and oil layers.

Homework Prepare for the final presentation. One group invites teachers, the head teacher, and if possible elders, parents and learners from other grades to the Magic Show.

Concept Built

Testing, peer feedback, and revision turn a science idea into a reliable, audience-ready demonstration.

Culminating Showcase

Showcase	The Science Magic Show — Final Presentation

Objective	Each group performs its science magic trick and explains the science behind it to a real audience beyond the teacher, then reflects on their learning.
Teacher's Note	Indoor. Combined classroom showcase, set up like Hornbill Festival stalls — it can spill into the corridor or yard if stalls need more space.
Duration	40 minutes (30 minutes presentation + 10 minutes reflection)

Materials Needed

Each group’s trick materials and posters, and a space set up as stalls for the show.

Activity Steps

1. Set up the room like stalls at the Hornbill Festival, one for each group. Learners present their tricks to classmates, teachers and the head teacher, and, where possible, elders, parents and learners from other grades.
2. As the audience visits each stall, the group shows the magic trick and explains the science behind it using their poster.
3. After each performance, the group asks the audience for two things they liked about the project and one thing that could be improved.
4. Independent reflection: learners think about and share in their group what they learned that they did not know before, what went well for their group, what challenge they faced and how they got past it, how their project helped answer the leading question, and one thing they would change or improve in their work.

Concept Built

Independent application of solute/solvent/solution ideas, saturation and solubility, and density in one integrated, locally-grounded science demonstration, explained confidently to a real audience.

By the end of this project, teachers together with students are required to design and perform at least one innovative science magic trick — showcased within the classroom or school, like a stall at the Hornbill Festival — that makes solutes, solvents, solutions or density visible and explainable to a real audience.

Reflect on Students' Understanding

Observe 5–6 learners each week during the tasks above. No written test is needed — watch what learners do with glasses, salt, and floating objects, and listen to how they explain their thinking.

Student Name	Identifies Solute & Solvent	Explains Saturation & Solubility	Explains Density (Float/Sink)	Explains the Trick's Science	Needs Support
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐
	☐	☐	☐	☐	☐

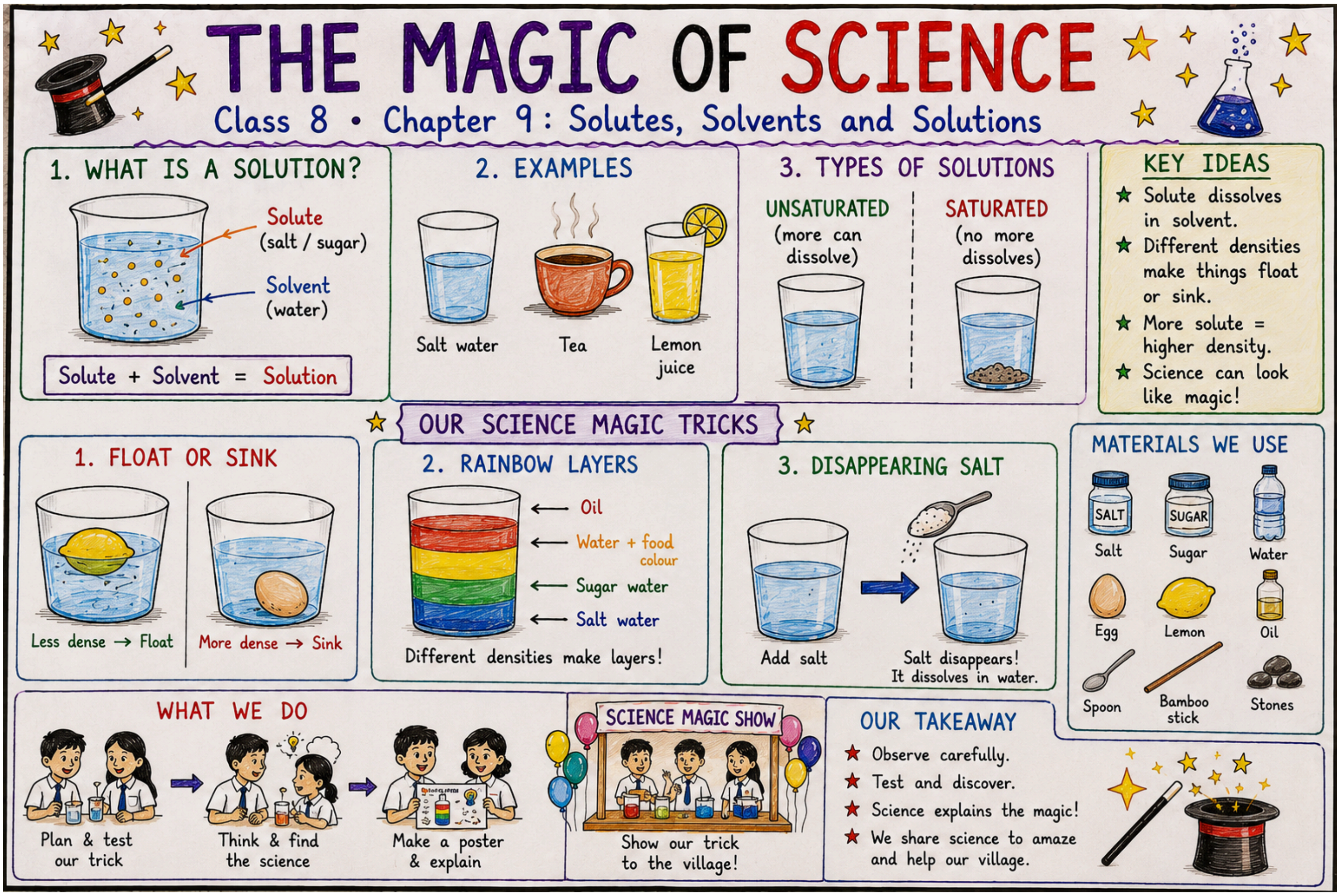
Scale (for each column): ☐ Not Yet ☐ Sometimes ☐ Consistently

Appendix 1: Peer Feedback Form

Give this to the group that watches your trick on Day 4.

Question	Group's response
Was the trick clear and easy to follow?	
Did the group explain the science correctly?	
Did the group work well together while performing?	
One specific suggestion to improve the trick or the explanation:	

Picture Guide



THE MAGIC OF SCIENCE
Class 8 • Chapter 9 : Solutes, Solvents and Solutions
1. WHAT IS A SOLUTION?
Solute
(salt / sugar)
Solvent
(water)
Solute + Solvent = Solution
2. EXAMPLES
Salt water
Tea
Lemon juice
3. TYPES OF SOLUTIONS
UNSATURATED
(more can dissolve)
SATURATED
(no more dissolves)
KEY IDEAS
Solute dissolves in solvent.
Different densities make things float or sink.
More solute = higher density.
Science can look like magic!
OUR SCIENCE MAGIC TRICKS
1. FLOAT OR SINK
Less dense → Float
More dense → Sink
2. RAINBOW LAYERS
Oil
Water + food colour
Sugar water
Salt water
Different densities make layers!
3. DISAPPEARING SALT
Add salt
Salt disappears!
It dissolves in water.
MATERIALS WE USE
SALT
Salt
SUGAR
Sugar
Water
Egg
Lemon
Oil
Spoon
Bamboo stick
Stones
WHAT WE DO
Plan & test our trick
Think & find the science
Make a poster & explain
SCIENCE MAGIC SHOW
Show our trick to the village!
OUR TAKEAWAY
Observe carefully.
Test and discover.
Science explains the magic!
We share science to amaze and help our village.